One Configuration, Tuned per Dataset

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The configuration. One model for all 8 datasets: inside and outside edges get two separate colors (their own weights in the GNN), trained with GLASS's recipe on GLASS's splits. On the synthetic datasets every shared node is replaced by one copy per subgraph. The real datasets run without copies, because their copies graphs do not fit in GPU memory.

What we ran, per dataset:

1. An Optuna search over aggregation, pooling, hidden size, layers, lr, dropouts, batch size, weight decay, norm, jumping knowledge, head, and (synthetic only) edge counts. Each trial is 3 seeds, scored by mean validation. Trials: hpo_neuro 80, hpo_metab and em_user 160, the rest 300. Weak trials were stopped after seed 0.

2. The 3 best trials on 10 seeds. The one with the best 10-seed validation is reported. Test was never used to choose.

	ours			published	
		copies			
dataset	one config	free coloring	one coloring	GLASS	GPEN
ppi_bp	55.3 ±0.9	58.6 ±2.2	60.6 ±2.3	61.9 ±2.2	**64.4** ±0.9
hpo_metab	60.5 ±2.8	**65.0** ±2.1	**65.0** ±2.1	61.4 ±1.6	63.9 ±0.9
hpo_neuro	64.6 ±0.7	68.4 ±0.5	67.6 ±1.1	68.5 ±1.6	**69.1** ±0.6
em_user	82.9 ±7.3	81.2 ±3.2	69.0 ±6.7	88.8 ±1.9	**91.2** ±1.3
density	**98.9** ±1.3	–	**98.9** ±1.3	93.0 ±2.8	95.6 ±1.6
cut_ratio	**95.2** ±2.0	–	**95.2** ±2.0	93.5 ±1.9	93.6 ±1.6
coreness	86.7 ±5.2	–	86.7 ±5.2	84.0 ±2.8	**87.6** ±1.9
component	**100.0** ±0.0	–	**100.0** ±0.0	**100.0** ±0.0	**100.0** ±0.0

em_user:
- Copies, free coloring: inside/outside plus the clique, with copies.
- Copies, one coloring: inside/outside with copies.

Table 1: Test micro-F1 (%), mean ± std over 10 seeds. One config: the search above (real datasets without copies). Copies, free coloring: the real datasets searched again with copies, the coloring picked by the search per dataset (attention left out). Copies, one coloring: inside / outside with copies everywhere; on the real datasets, the free-coloring winner's settings with the coloring switched, so on hpo_metab (which picked it) and on the synthetic datasets it equals the column before. GLASS and GPEN: published numbers from the GPEN paper. Bold: the highest score in the row.

1. Overfitting. Train F1 is about 99.8 on both em_user and ppi_bp, against 89 on hpo_metab, where we beat the baselines.
2. Noisy validation. Both have small validation sets (em_user has about 65 subgraphs), so the search picks winners by luck. ppi_bp's winner scores 69.6 on validation but 58.6 on test, and on em_user validation and test barely track each other (0.22 correlation).